

Netflix Data Analysis

Report

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1. Project Overview

Objective

The objective of this project was to perform **Exploratory Data Analysis (EDA)** on the Netflix Movies and TV Shows dataset using **Python (Pandas)** in Google Colab. The goal was to understand the dataset, improve its quality through data cleaning, identify content trends, and extract meaningful insights.

2. Dataset Overview

We first explored the dataset to understand its structure.

Dataset Summary

Attribute	Value
Total Records	8,807
Total Columns	12
Content Types	Movies and TV Shows

The dataset includes information such as:

- Show ID
- Type
- Title
- Director
- Cast
- Country

- Date Added
- Release Year
- Rating
- Duration
- Genre (listed_in)
- Description

3. Data Cleaning

Before analysis, we examined data quality.

Missing Values Found

Several columns contained missing values.

The most affected columns were:

- Director
- Cast
- Country

A few missing values also existed in:

- Date Added
- Rating
- Duration

Cleaning Strategy

Instead of deleting a large number of records, we used different strategies depending on the importance of each column.

- Missing values in **Director**, **Cast**, and **Country** were replaced with "Unknown" to preserve records.
- Records missing **Date Added**, **Rating**, or **Duration** were removed because only a very small number of entries were affected.

This approach minimized data loss while improving data quality.

4. Exploratory Data Analysis

A. Movies vs TV Shows

Analysis showed that:

- Movies represent the majority of Netflix's content.
- TV Shows form a smaller but significant portion of the catalog.

Insight

Netflix's library in this dataset is more movie-oriented than TV Show-oriented.

B. Decade-wise Analysis

A new **Decade** column was created from the release year.

The analysis revealed:

- Very little content before the 1980s.
- Rapid growth after 2000.
- The 2010s contain the highest number of titles.

Insight

The 2010s were the most content-rich decade in the dataset, with Movies significantly outnumbering TV Shows, although TV Shows also experienced considerable growth.

C. Country-wise Analysis

Country analysis identified the major contributors to Netflix's catalog.

Key observations:

- United States and India are among the leading content-producing countries.
- Content is distributed across many countries, reflecting Netflix's global reach.

Insight

Netflix maintains a geographically diverse content library rather than relying on a single country.

D. Genre Analysis

Since a title can belong to multiple genres, the `listed_in` column was transformed so that each

genre could be analyzed individually.

Top genres included:

- International Movies
- Dramas
- Comedies
- International TV Shows
- Documentaries

Insight

International content represents a substantial portion of the Netflix catalog, highlighting the platform's focus on serving audiences across different regions.

E. Cast Analysis

The Cast column was also transformed to analyze individual actors.

Among recorded actor names:

- Anupam Kher
- Shah Rukh Khan
- Naseeruddin Shah
- Akshay Kumar

appeared frequently.

A large number of records contained missing cast information, which had been replaced with "Unknown" during data cleaning.

Insight

The frequent appearance of Indian actors aligns with the significant amount of Indian content observed in the dataset.

5. Key Learnings

This project strengthened my understanding of:

- Data cleaning techniques

- Handling missing values
- Filtering and grouping data
- Feature engineering
- Frequency analysis
- Multi-value columns
- Exploratory Data Analysis (EDA)
- Analytical thinking

More importantly, I learned that successful data analysis is not only about writing code but also about asking meaningful questions and interpreting results responsibly.

6. Limitations

Every dataset has limitations.

This analysis should be interpreted while considering the following:

- Some metadata (Director, Cast, Country) was incomplete.
- The dataset represents Netflix titles available at the time of collection and may not reflect the current catalog.
- Genre counts exceed the number of titles because a single title can belong to multiple genres.
- This project focused on descriptive analysis and did not include predictive modeling.

7. Conclusion

The Netflix dataset provides valuable insights into the platform's content distribution.

The analysis suggests that:

- Movies dominate the catalog.
- The largest growth occurred during the 2010s.
- Netflix offers content from many countries.
- International genres occupy a major share of the catalog.
- Indian cinema has a strong representation in the dataset.

Overall, this project demonstrates how **Pandas** can be used to clean, transform, and analyze real-world datasets while developing the analytical mindset required in data analytics.